

北京大学期末试题解答

2021-2022学年第二学期

考试科目： 数学物理方法（下） 考试时间： 2022年6月14日

姓 名： 学 号：

本试题共 五 道大题，满分 100 分

注意：各题均需写出必要的关键步骤。

- 一、(20分) 一圆柱体底面半径为 a ，高为 h ，其中 a 和 h 均为已知常数。上底面和圆柱侧面均保持温度为零度，下底面以垂直于底面的恒定热流密度 q 输入热量，其中 q 为已知常矢量。求达到平衡后圆柱体内温度的稳定分布。

Solution: 以圆柱体的下底面为 x - y 平面，柱轴为 z 轴建立坐标架。设 t 时刻圆柱体内 (x, y, z) 处的温度为 $u(t, x, y, z)$ ，达到平衡后 $u(x, y, z)$ 满足定解问题

$$\begin{cases} \nabla^2 u(x, y, z) = 0, & x^2 + y^2 < a^2, 0 < z < h \\ u|_{x^2+y^2=a^2} = 0, & 0 \leq z \leq h \\ -k \frac{\partial u}{\partial z} \Big|_{z=0} = q, & u|_{z=h} = 0, \quad x^2 + y^2 \leq a^2 \end{cases}$$

在柱坐标系下，由于方程和所有边界条件均与 ϕ 无关，因此 u 与 ϕ 无关。 $u(\rho, z)$ 满足定解问题

$$\begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{\partial^2 u}{\partial z^2} = 0, & 0 < \rho < a, 0 < z < h \\ u|_{\rho=0} \text{ 有界}, & u|_{\rho=a} = 0, \quad 0 \leq z \leq h \\ \frac{\partial u}{\partial z} \Big|_{z=0} = -\frac{q}{k}, & u|_{z=h} = 0, \quad 0 \leq \rho \leq a \end{cases}$$

方法一：以 ρ 方向构成本征值问题

一般解为

$$u(\rho, z) = \sum_{n=1}^{\infty} A_n \frac{\sinh \frac{\mu_n}{a} (h-z)}{\cosh \frac{\mu_n}{a} h} J_0 \left(\frac{\mu_n}{a} \rho \right)$$

其中 μ_n 是 $J_0(x)$ 的第 n 个正零点, $n = 1, 2, \dots$, 模方

$$\int_0^a J_0^2 \left(\frac{\mu_n}{a} \rho \right) \rho d\rho = \frac{a^2}{2} [J_0'(\mu_n)]^2$$

$$\left. \frac{\partial u}{\partial z} \right|_{z=0} = - \sum_{n=1}^{\infty} A_n \frac{\mu_n}{a} J_0 \left(\frac{\mu_n}{a} \rho \right) = -\frac{q}{k}$$

$$\Rightarrow A_n = \frac{\int_0^a \frac{q}{k} J_0 \left(\frac{\mu_n}{a} \rho \right) \rho d\rho}{\frac{\mu_n}{a} \int_0^a J_0^2 \left(\frac{\mu_n}{a} \rho \right) \rho d\rho} = \frac{2qa}{k\mu_n^2 J_1(\mu_n)}$$

$$u(\rho, z) = \frac{2qa}{k} \sum_{n=1}^{\infty} \frac{1}{\mu_n^2 J_1(\mu_n)} \frac{\sinh \frac{\mu_n}{a} (h-z)}{\cosh \frac{\mu_n}{a} h} J_0 \left(\frac{\mu_n}{a} \rho \right)$$

方法二：以 z 方向构成本征值问题

令

$$u(\rho, z) = \frac{q}{k}(h-z) + v(\rho, z)$$

则 $v(\rho, z)$ 满足定解问题

$$\begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial v}{\partial \rho} \right) + \frac{\partial^2 v}{\partial z^2} = 0, & 0 < \rho < a, 0 < z < h \\ v|_{\rho=0} \text{ 有界}, & v|_{\rho=a} = \frac{q}{k}(z-h), & 0 \leq z \leq h \\ \left. \frac{\partial v}{\partial z} \right|_{z=0} = 0, & v|_{z=h} = 0, & 0 \leq \rho \leq a \end{cases}$$

一般解为

$$v(\rho, z) = \sum_{n=0}^{\infty} B_n \cos \frac{(2n+1)\pi}{2h} z I_0 \left(\frac{(2n+1)\pi}{2h} \rho \right)$$

$$v|_{\rho=a} = \sum_{n=0}^{\infty} B_n \cos \frac{(2n+1)\pi}{2h} z I_0 \left(\frac{(2n+1)\pi}{2h} a \right) = \frac{q}{k}(z-h)$$

$$\begin{aligned}\Rightarrow B_n &= \frac{1}{I_0\left(\frac{(2n+1)\pi}{2h}a\right)} \frac{2}{h} \int_0^h \frac{q}{k}(z-h) \cos \frac{(2n+1)\pi}{2h}z dz \\ &= -\frac{1}{I_0\left(\frac{(2n+1)\pi}{2h}a\right)} \frac{2q}{hk} \left[\frac{2h}{(2n+1)\pi} \right]^2\end{aligned}$$

所以原定解问题的解为

$$u(\rho, z) = \frac{q}{k}(h-z) - \frac{8qh}{\pi^2 k} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2 I_0\left(\frac{(2n+1)\pi}{2h}a\right)} \cos \frac{(2n+1)\pi}{2h}z I_0\left(\frac{(2n+1)\pi}{2h}\rho\right)$$

二、 (25分) 求解球内振动问题

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \nabla^2 u = v_0 z, & x^2 + y^2 + z^2 < r_0^2, t > 0 \\ \frac{\partial u}{\partial n} \Big|_{x^2+y^2+z^2=r_0^2} = 0, & t \geq 0 \\ u|_{t=0} = 0, \quad \frac{\partial u}{\partial t} \Big|_{t=0} = 0, & x^2 + y^2 + z^2 \leq r_0^2 \end{cases}$$

其中 $a > 0$, $v_0 > 0$ 和 $r_0 > 0$ 都是已知常数。

Solution: 在球坐标系下, 由于方程和所有定解条件均与 ϕ 无关, 因此 u 与 ϕ 无关。
 $u(r, \theta, t)$ 满足定解问题

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) \right] = v_0 r \cos \theta, & 0 < r < r_0, 0 < \theta < \pi, t > 0 \\ u|_{r=0} \text{ 有界}, \quad \frac{\partial u}{\partial r} \Big|_{r=r_0} = 0, & 0 \leq \theta \leq \pi, t \geq 0 \\ u|_{\theta=0} \text{ 有界}, \quad u|_{\theta=\pi} \text{ 有界}, & 0 \leq r \leq r_0, t \geq 0 \\ u|_{t=0} = 0, \quad \frac{\partial u}{\partial t} \Big|_{t=0} = 0, & 0 \leq r \leq r_0, 0 \leq \theta \leq \pi \end{cases}$$

令

$$u(r, \theta, t) = v(r, t) \cos \theta$$

则 $v(r, t)$ 满足定解问题

$$\begin{cases} \frac{\partial^2 v}{\partial t^2} - a^2 \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v}{\partial r} \right) - \frac{2}{r^2} v \right] = v_0 r, & 0 < r < r_0, t > 0 \\ v|_{r=0} \text{ 有界}, \quad \frac{\partial v}{\partial r} \Big|_{r=r_0} = 0, & t \geq 0 \\ v|_{t=0} = 0, \quad \frac{\partial v}{\partial t} \Big|_{t=0} = 0, & 0 \leq r \leq r_0 \end{cases}$$

令

$$v(r, t) = R(r)T(t)$$

相应齐次方程分离变量，得本征值问题

$$\begin{cases} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR(r)}{dr} \right) + \left(\lambda - \frac{2}{r^2} \right) R(r) = 0, & 0 < r < r_0 \\ R(0) \text{有界}, & R'(r_0) = 0, \end{cases}$$

容易证得

$$\lambda \geq 0$$

又易知 $\lambda = 0$ 不是本征值，因此有

$$\lambda > 0$$

本征方程的通解为

$$R(r) = A j_1(\sqrt{\lambda}r) + B n_1(\sqrt{\lambda}r)$$

$$R(0) \text{有界} \Rightarrow B = 0$$

$$R'(r_0) = 0 \Rightarrow \left. \frac{d}{dr} j_1(\sqrt{\lambda}r) \right|_{r=r_0} = 0 \Rightarrow \lambda_n = \left(\frac{\mu_n}{r_0} \right)^2, n = 1, 2, \dots$$

其中 μ_n 是方程 $j_1'(x) = 0$ 的第 n 个正根。本征值问题的解为

$$\begin{cases} \lambda_n = \left(\frac{\mu_n}{r_0} \right)^2, \\ R_n(r) = j_1 \left(\frac{\mu_n}{r_0} r \right), \end{cases} \quad n = 1, 2, \dots$$

求本征函数的模方

$$\begin{aligned} \int_0^{r_0} R_n(r) R_n(r) r^2 dr &= \lim_{\lambda \rightarrow \lambda_n} \int_0^a R(r) R_n(r) r^2 dr \\ &= \lim_{\lambda \rightarrow \lambda_n} \frac{r^2 \left(R(r) \frac{dR_n(r)}{dr} - R_n(r) \frac{dR(r)}{dr} \right) \Big|_{r=0}^{r=r_0}}{\lambda - \lambda_n} \\ &= - \lim_{\lambda \rightarrow \lambda_n} r_0^2 R_n(r_0) \frac{d}{d\lambda} \frac{dR(r)}{dr} \Big|_{r=r_0} \\ &= - \lim_{\lambda \rightarrow \lambda_n} r_0^2 R_n(r_0) \frac{d}{d\lambda} \frac{dj_1(\sqrt{\lambda}r)}{dr} \Big|_{r=r_0} = - \frac{r_0^3}{2} j_1(\mu_n) j_1''(\mu_n) \end{aligned}$$

$v(r, t)$ 用相应齐次方程的本征函数展开

$$v(r, t) = \sum_{n=1}^{\infty} T_n(t) j_1 \left(\frac{\mu_n}{r_0} r \right)$$

非齐次项 $v_0 r$ 也用本征函数组 $\{R_n(r)\}$ 展开

$$v_0 r = \sum_{n=1}^{\infty} c_n j_1 \left(\frac{\mu_n}{r_0} r \right)$$

其中

$$c_n = \frac{\int_0^{r_0} v_0 r R_n(r) r^2 dr}{\int_0^{r_0} R_n(r) R_n(r) r^2 dr} = -\frac{v_0}{\frac{r_0^3}{2} j_1(\mu_n) j_1''(\mu_n)} \left(\frac{r_0}{\mu_n} \right)^4 \int_0^{\mu_n} x^3 j_1(x) dx = -\frac{2v_0 r_0 j_2(\mu_n)}{j_1(\mu_n) j_1''(\mu_n)}$$

得 $T_n(t)$ 满足定解问题

$$\begin{cases} T_n''(t) + \left(\frac{\mu_n a}{r_0} \right)^2 T_n(t) = -\frac{2v_0 r_0 j_2(\mu_n)}{j_1(\mu_n) j_1''(\mu_n)} \\ T_n(0) = 0, \quad T_n'(0) = 0 \end{cases}$$

解得

$$T_n(t) = \frac{2v_0 r_0^3 j_2(\mu_n)}{\mu_n^2 a^2 j_1(\mu_n) j_1''(\mu_n)} \left(\cos \frac{\mu_n a t}{r_0} - 1 \right)$$

因此原定解问题的解为

$$\begin{aligned} u(r, \theta, t) &= \cos \theta \sum_{n=1}^{\infty} \frac{2v_0 r_0^3 j_2(\mu_n)}{\mu_n^2 a^2 j_1(\mu_n) j_1''(\mu_n)} \left(\cos \frac{\mu_n a t}{r_0} - 1 \right) j_1 \left(\frac{\mu_n}{r_0} r \right) \\ &= \cos \theta \sum_{n=1}^{\infty} \frac{2v_0 r_0^3 j_2(\mu_n)}{\mu_n^2 a^2 j_1(\mu_n) j_1(\mu_n) \left(1 - \frac{2}{\mu_n^2} \right)} \left(1 - \cos \frac{\mu_n a t}{r_0} \right) j_1 \left(\frac{\mu_n}{r_0} r \right) \end{aligned}$$

其中 μ_n 是方程 $j_1'(x) = 0$ 的第 n 个正根。

三、 (20分) 一维Dirac Hamiltonian算符

$$\hat{H} = \begin{pmatrix} -i \frac{d}{dx} & m_1(x) - i m_2(x) \\ m_1(x) + i m_2(x) & i \frac{d}{dx} \end{pmatrix}$$

其中 $a \leq x \leq b$, $m_1(x)$ 和 $m_2(x)$ 都是实函数。算符 $\hat{H}$ 作用在两分量的自旋子

$$F(x) = \begin{pmatrix} f_1(x) \\ f_2(x) \end{pmatrix}$$

上, 其中 $f_1(x)$ 和 $f_2(x)$ 都是复值函数。内积定义为

$$\langle F | G \rangle = \int_a^b F^\dagger(x) G(x) dx = \int_a^b [f_1^*(x) g_1(x) + f_2^*(x) g_2(x)] dx$$

其中上标*表示复共轭,

$$F^\dagger(x) = (f_1^*(x), f_2^*(x)), \quad G(x) = \begin{pmatrix} g_1(x) \\ g_2(x) \end{pmatrix}$$

(1) 什么条件下算符 $\hat{H}$ 有伴算符?

(2) 分别给出以下两种情形的例子:

情形一 有伴算符但不是自伴算符的 $\hat{H}$ 及其伴算符 $\hat{H}^\dagger$;

情形二 自伴算符 $\hat{H}$ 。

Solution: $\forall F(x) \in D(\hat{H}),$ 对 $G(x)$

$$\begin{aligned} \langle G | \hat{H} F \rangle &= \int_a^b G^\dagger \hat{H} F dx \\ &= \int_a^b \left\{ g_1^*(x) \left[-i \frac{df_1}{dx} + (m_1(x) - im_2(x)) f_2(x) \right] \right. \\ &\quad \left. + g_2^*(x) \left[(m_1(x) + im_2(x)) f_1(x) + i \frac{df_2}{dx} \right] \right\} dx \\ &= i [-g_1^*(x) f_1(x) + g_2^*(x) f_2(x)]_{x=a}^{x=b} \\ &\quad + \int_a^b \left\{ \left[-i \frac{dg_1}{dx} + g_2(x)(m_1(x) - im_2(x)) \right]^* f_1(x) \right. \\ &\quad \left. + \left[g_1(x)(m_1(x) + im_2(x)) + i \frac{dg_2}{dx} \right]^* f_2(x) \right\} dx \end{aligned}$$

因此

(1) 当

$$g_2^*(b) f_2(b) - g_1^*(b) f_1(b) = g_2^*(a) f_2(a) - g_1^*(a) f_1(a)$$

时, $\hat{H}$ 有伴算符。

(2) 分两种情形举例:

情形一 例如当 $F(x)$ 满足

$$f_1(a) = 2f_1(b) \text{ 且 } f_2(a) = 2f_2(b)$$

时，即

$$F(a) = 2F(b)$$

时，要求 $G(x)$ 满足

$$2g_1(a) = g_1(b) \text{ 且 } 2g_2(a) = g_2(b)$$

即

$$2G(a) = G(b)$$

此时

$$\left\{ \begin{array}{l} \hat{H} = \begin{pmatrix} -i\frac{d}{dx} & m_1(x) - im_2(x) \\ m_1(x) + im_2(x) & i\frac{d}{dx} \end{pmatrix} \\ D(\hat{H}) = \left\{ F(x) = \begin{pmatrix} f_1(x) \\ f_2(x) \end{pmatrix} : \begin{array}{l} a \leq x \leq b, f_1(x) \in L^2[a, b], \\ f_2(x) \in L^2[a, b], F(a) = 2F(b) \end{array} \right\} \end{array} \right\}$$

有伴算符

$$\left\{ \begin{array}{l} \hat{H}^\dagger = \begin{pmatrix} -i\frac{d}{dx} & m_1(x) - im_2(x) \\ m_1(x) + im_2(x) & i\frac{d}{dx} \end{pmatrix} \\ D(\hat{H}^\dagger) = \left\{ G(x) = \begin{pmatrix} g_1(x) \\ g_2(x) \end{pmatrix} : \begin{array}{l} a \leq x \leq b, g_1(x) \in L^2[a, b], \\ g_2(x) \in L^2[a, b], 2G(a) = G(b) \end{array} \right\} \end{array} \right\}$$

$$\hat{H}^\dagger \neq \hat{H}$$

情形二 例如当 $F(x)$ 满足

$$f_1(a) = f_1(b) \text{ 且 } f_2(a) = f_2(b)$$

时，即

$$F(a) = F(b)$$

时，要求 $G(x)$ 也满足

$$g_1(a) = g_1(b) \text{ 且 } g_2(a) = g_2(b)$$

即

$$G(a) = G(b)$$

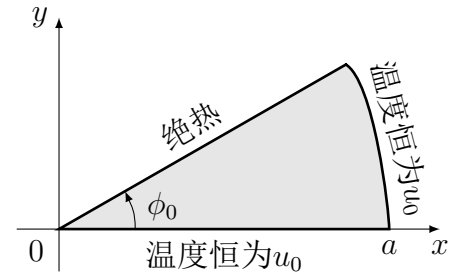
此时

$$\left\{ \begin{array}{l} \hat{H} = \begin{pmatrix} -i\frac{d}{dx} & m_1(x) - im_2(x) \\ m_1(x) + im_2(x) & i\frac{d}{dx} \end{pmatrix} \\ D(\hat{H}) = \left\{ F(x) = \begin{pmatrix} f_1(x) \\ f_2(x) \end{pmatrix} : \begin{array}{l} a \leq x \leq b, f_1(x) \in L^2[a, b], \\ f_2(x) \in L^2[a, b], F(a) = F(b) \end{array} \right\} \end{array} \right\}$$

是自伴算符, 即

$$\hat{H}^\dagger = \hat{H}$$

四、(20分) 一高度为无穷 ($-\infty < z < \infty$) 的柱体, 横截面为如右图所示的扇形, 其张角为 ϕ_0 , 半径为 a , 其中 ϕ_0 和 a 均为已知常数。柱体一个侧平面绝热, 另一个侧平面及圆弧形侧面上温度均恒为 u_0 , 其中 $u_0 \neq 0$ 为已知常数。柱体初始温度为 $u|_{t=0} = f(x, y, z)$, 其中 f 为已知实函数。求 $t > 0$ 时柱体的温度分布。



第4.题无穷高柱体的横截面

Solution: 设 t 时刻柱体 (x, y, z) 处的温度为 $u(x, y, z, t)$, 选取柱坐标系, 则 u 满足定解问题

$$\left\{ \begin{array}{l} \frac{\partial u}{\partial t} - \kappa \left[\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2} \right] = 0, \quad 0 < \rho < a, 0 < \phi < \phi_0, -\infty < z < \infty, t > 0 \\ u|_{\rho=0} \text{ 有界}, \quad u|_{\rho=a} = u_0, \quad 0 \leq \phi \leq \phi_0, -\infty < z < \infty, t \geq 0 \\ u|_{\phi=0} = u_0, \quad \frac{\partial u}{\partial \phi} \Big|_{\phi=\phi_0} = 0, \quad 0 \leq \rho \leq a, -\infty < z < \infty, t \geq 0 \\ u|_{z \rightarrow \pm\infty} \text{ 有界}, \quad 0 \leq \rho \leq a, 0 \leq \phi \leq \phi_0, t \geq 0 \\ u|_{t=0} = f(\rho, \phi, z), \quad 0 \leq \rho \leq a, 0 \leq \phi \leq \phi_0, -\infty < z < \infty \end{array} \right.$$

令

$$u = v(\rho, \phi, z, t) + u_0$$

则 v 满足定解问题

$$\left\{ \begin{array}{ll} \frac{\partial v}{\partial t} - \kappa \left[\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial v}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 v}{\partial \phi^2} + \frac{\partial^2 v}{\partial z^2} \right] = 0, & 0 < \rho < a, 0 < \phi < \phi_0, -\infty < z < \infty, t > 0 \\ v|_{\rho=0} \text{ 有界}, & v|_{\rho=a} = 0, & 0 \leq \phi \leq \phi_0, -\infty < z < \infty, t \geq 0 \\ v|_{\phi=0} = 0, & \frac{\partial v}{\partial \phi} \Big|_{\phi=\phi_0} = 0, & 0 \leq \rho \leq a, -\infty < z < \infty, t \geq 0 \\ v|_{z \rightarrow \pm \infty} \text{ 有界}, & & 0 \leq \rho \leq a, 0 \leq \phi \leq \phi_0, t \geq 0 \\ v|_{t=0} = f(\rho, \phi, z) - u_0, & & 0 \leq \rho \leq a, 0 \leq \phi \leq \phi_0, -\infty < z < \infty \end{array} \right.$$

把 v 按照 ϕ 方向的本征函数展开, 设

$$v(\rho, \phi, z, t) = \sum_{n=1}^{\infty} w_n(\rho, z, t) \sin \frac{(2n+1)\pi}{2\phi_0} \phi$$

则 $w_n(\rho, z, t)$ 满足定解问题

$$\left\{ \begin{array}{ll} \frac{\partial w_n}{\partial t} - \kappa \left\{ \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial w_n}{\partial \rho} \right) - \left[\frac{(2n+1)\pi}{2\phi_0} \right]^2 \frac{w_n}{\rho^2} + \frac{\partial^2 w_n}{\partial z^2} \right\} = 0, & \rho > 0, -\infty < z < \infty, t > 0 \\ w_n|_{\rho=0} \text{ 有界}, & w_n|_{\rho=a} = 0, & -\infty < z < \infty, t \geq 0 \\ w_n|_{t=0} = f_n(\rho, z), & & \rho \geq 0, -\infty < z < \infty \end{array} \right.$$

其中

$$f_n(\rho, z) = \frac{2}{\phi_0} \int_0^{\phi_0} (f(\rho, \phi', z) - u_0) \sin \frac{(2n+1)\pi}{2\phi_0} \phi' d\phi' \quad (1)$$

对 z 作Fourier变换, 记

$$\begin{aligned} \mathcal{F} \{w_n(\rho, z, t)\} &= W_n(\rho, k, t) \\ \mathcal{F} \{f_n(\rho, z)\} &= F_n(\rho, k) \end{aligned}$$

得

$$\left\{ \begin{array}{ll} \frac{\partial W_n}{\partial t} - \kappa \left\{ \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial W_n}{\partial \rho} \right) - \left[\frac{(2n+1)\pi}{2\phi_0} \right]^2 \frac{W_n}{\rho^2} - k^2 W_n \right\} = 0, & \rho > 0, t > 0 \\ W_n|_{\rho=0} \text{ 有界}, & W_n|_{\rho=a} = 0, & t \geq 0 \\ W_n|_{t=0} = F_n(\rho, k), & & 0 \leq \rho \leq a, \end{array} \right.$$

设

$$W_n = e^{-\kappa k^2 t} V_n(\rho, k, t)$$

则 $V_n(\rho, k, t)$ 满足定解问题

$$\begin{cases} \frac{\partial V_n}{\partial t} - \kappa \left\{ \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V_n}{\partial \rho} \right) - \left[\frac{(2n+1)\pi}{2\phi_0} \right]^2 \frac{V_n}{\rho^2} \right\} = 0, & \rho > 0, t > 0 \\ V_n|_{\rho=0} \text{ 有界}, & V_n|_{\rho=a} = 0, & t \geq 0 \\ V_n|_{t=0} = F_n(\rho, k), & & 0 \leq \rho \leq a, \end{cases}$$

解得

$$V_n(\rho, t) = \sum_{m=1}^{\infty} A_m e^{-\kappa \left(\frac{\mu_m^n}{a} \right)^2 t} J_{\frac{(2n+1)\pi}{2\phi_0}} \left(\frac{\mu_m^n}{a} \rho \right)$$

其中 μ_m^n 是 $J_{\frac{(2n+1)\pi}{2\phi_0}}(x)$ 的第 m 个正零点, $m = 1, 2, \dots$,

$$A_m = \frac{\int_0^a F_n(\rho, k) J_{\frac{(2n+1)\pi}{2\phi_0}} \left(\frac{\mu_m^n}{a} \rho \right) \rho d\rho}{\int_0^a \left[J_{\frac{(2n+1)\pi}{2\phi_0}} \left(\frac{\mu_m^n}{a} \rho \right) \right]^2 \rho d\rho}$$

其中模方

$$\int_0^a \left[J_{\frac{(2n+1)\pi}{2\phi_0}} \left(\frac{\mu_m^n}{a} \rho \right) \right]^2 \rho d\rho = \frac{a^2}{2} \left[J'_{\frac{(2n+1)\pi}{2\phi_0}}(\mu_m^n) \right]^2$$

因此

$$W_n(\rho, k, t) = e^{-\kappa k^2 t} \sum_{m=1}^{\infty} \frac{\int_0^a F_n(\rho', k) J_{\frac{(2n+1)\pi}{2\phi_0}} \left(\frac{\mu_m^n}{a} \rho' \right) \rho' d\rho'}{\frac{a^2}{2} \left[J'_{\frac{(2n+1)\pi}{2\phi_0}}(\mu_m^n) \right]^2} e^{-\kappa \left(\frac{\mu_m^n}{a} \right)^2 t} J_{\frac{(2n+1)\pi}{2\phi_0}} \left(\frac{\mu_m^n}{a} \rho \right)$$

利用

$$\mathcal{F}^{-1} \left\{ e^{-\kappa k^2 t} \right\} = \frac{1}{\sqrt{2\kappa t}} \exp \left[-\frac{z^2}{4\kappa t} \right]$$

以及卷积, 得

$$\mathcal{F}^{-1} \left\{ e^{-\kappa k^2 t} F_n(\rho', k) \right\} = \int_{-\infty}^{\infty} f_n(\rho', \zeta) \frac{1}{\sqrt{2\kappa t}} \exp \left[-\frac{(z - \zeta)^2}{4\kappa t} \right] d\zeta$$

因此

$$\begin{aligned} w_n(\rho, z, t) &= \frac{1}{\sqrt{2\kappa t}} \sum_{m=1}^{\infty} \frac{e^{-\kappa \left(\frac{\mu_m^n}{a} \right)^2 t} J_{\frac{(2n+1)\pi}{2\phi_0}} \left(\frac{\mu_m^n}{a} \rho \right)}{\frac{a^2}{2} \left[J'_{\frac{(2n+1)\pi}{2\phi_0}}(\mu_m^n) \right]^2} \\ &\quad \times \int_0^a \int_{-\infty}^{\infty} f_n(\rho', \zeta) \exp \left[-\frac{(z - \zeta)^2}{4\kappa t} \right] d\zeta J_{\frac{(2n+1)\pi}{2\phi_0}} \left(\frac{\mu_m^n}{a} \rho' \right) \rho' d\rho' \end{aligned}$$

原定解问题的解为

$$\begin{aligned}
 u(\rho, \phi, z, t) = & u_0 + \frac{2}{a^2 \phi_0} \sqrt{\frac{2}{\kappa t}} \sum_{n=1}^{\infty} \sin \frac{(2n+1)\pi}{2\phi_0} \phi \sum_{m=1}^{\infty} \frac{e^{-\kappa \left(\frac{\mu_m^n}{a}\right)^2 t} J_{\frac{(2n+1)\pi}{2\phi_0}} \left(\frac{\mu_m^n}{a} \rho\right)}{\left[J'_{\frac{(2n+1)\pi}{2\phi_0}}(\mu_m^n)\right]^2} \\
 & \cdot \int_0^a \int_{-\infty}^{\infty} \int_0^{\phi_0} [f(\rho', \phi', \zeta) - u_0] \sin \frac{(2n+1)\pi}{2\phi_0} \phi' d\phi' \\
 & \cdot \exp \left[-\frac{(z-\zeta)^2}{4\kappa t} \right] d\zeta J_{\frac{(2n+1)\pi}{2\phi_0}} \left(\frac{\mu_m^n}{a} \rho'\right) \rho' d\rho'
 \end{aligned}$$

五、 (15分) 求下列定解问题相应的Green函数:

$$\begin{cases} \nabla^2 u + k^2 u = f(x, y, z), & -\infty < x < \infty, -\infty < y < \infty, z > 0 \\ u|_{z=0} = g(x, y), & -\infty < x < \infty, -\infty < y < \infty, \\ u \text{ 在无穷远只有会聚波,} & (\text{取时间因子为 } e^{-i\omega t}) \end{cases}$$

其中 $k > 0$ 是已知常数, f 和 g 都是已知函数。

Solution: 相应的Green函数满足定解问题

$$\begin{cases} \nabla^2 G(\mathbf{r}; \mathbf{r}') + k^2 G(\mathbf{r}; \mathbf{r}') = \delta(\mathbf{r} - \mathbf{r}'), & -\infty < x, x' < \infty, -\infty < y, y' < \infty, z, z' > 0 \\ G(\mathbf{r}; \mathbf{r}')|_{z=0} = 0, & -\infty < x, x' < \infty, -\infty < y, y' < \infty, z' > 0 \\ G(\mathbf{r}; \mathbf{r}') \text{ 在无穷远只有会聚波,} \end{cases}$$

已知函数

$$G_1 = -\frac{e^{-ik|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|}$$

满足方程

$$\nabla^2 G_1(\mathbf{r}; \mathbf{r}') + k^2 G_1(\mathbf{r}; \mathbf{r}') = \delta(\mathbf{r} - \mathbf{r}')$$

以及在无穷远只有会聚波, 令

$$G(\mathbf{r}; \mathbf{r}') = G_1(\mathbf{r}; \mathbf{r}') + v$$

则要求 v 满足

$$\left\{ \begin{array}{ll} \nabla^2 v + k^2 v = 0, & -\infty < x < \infty, -\infty < y < \infty, z > 0 \\ v|_{z=0} = -G_1(\mathbf{r}; \mathbf{r}')|_{z=0} = \frac{e^{-ik\sqrt{(x-x')^2+(y-y')^2+z'^2}}}{4\pi\sqrt{(x-x')^2+(y-y')^2+z'^2}}, & -\infty < x, x' < \infty, -\infty < y, y' < \infty, \\ v \text{ 在无穷远只有会聚波,} \end{array} \right.$$

因此解得

$$v = \frac{e^{-ik\sqrt{(x-x')^2+(y-y')^2+(z+z')^2}}}{4\pi\sqrt{(x-x')^2+(y-y')^2+(z+z')^2}}$$

所以原定解问题的解为

$$G(\mathbf{r}; \mathbf{r}') = \frac{e^{-ik\sqrt{(x-x')^2+(y-y')^2+(z+z')^2}}}{4\pi\sqrt{(x-x')^2+(y-y')^2+(z+z')^2}} - \frac{e^{-ik\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}}}{4\pi\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}},$$

其中

$$-\infty < x, x' < \infty, -\infty < y, y' < \infty, z, z' > 0$$